# Supplementary Information

Internal consistency, not plausibility, decides whether AI agents relay database errors

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## Supplementary Note 1: the first deployment run and its audit

The first full run of the deployment experiment began on 25 October 2026 under the author's own macOS account. Biomni's system prompt lists its data-lake directory, and that directory sat inside the study's repository, so the path shown to the model contained the strings `bioagent-plausibility` and `biomni_spike`. Biomni's code executor was not sandboxed: code the agent wrote could read the author's filesystem and reach the internet, and the study's repository was public.

After the run, every transcript was audited with three regular expressions, recorded in `results/analysis/audit_biomni_environment.py`: *awareness* (phrases inferring a plausibility benchmark, a test, or an evaluation of the agent), *repository pattern* (code executions referencing the author's home directory outside the data lake, the study's repository files, `git` commands or process listings) and *metadata vocabulary* (terms from the study's task files, such as `corrupted_value` or "answer key"). DeepSeek V4.1 Flash matched none of the first two in any run and the third in one run (u015), which caught the corruption regardless. Gemini 3.8 Flash matched awareness in 44 runs, the repository pattern in 98 and the vocabulary in 2. Aware runs behaved differently: Flash's aware runs relayed plausible values in 1 of 23 delivered runs, against 68 of 265 for the others. Awareness does not account for the difference between the runs, however: both models relayed plausible values less often in the rerun (Gemini 3.8 Flash 21% against 24%; DeepSeek, which showed no awareness, 17% against 28%), which indicates run-to-run variation of about ten points in this setting. We report the rerun as registered, and both runs gave the same direction for the derivation-route hypothesis (odds ratios 5.7 and 17.2 for DeepSeek, 3.6 and 6.9 for Gemini 3.8 Flash; Supplementary Table 1). The external files Flash fetched were Biomni's own repository (its evaluation task list) and a third-party benchmark (Bio-Bench); no transcript shows the study's files being read.

The corrupted conditions were rerun on 3 November 2026 under a second macOS user with no read access to the author's home directory, with the data lake at a neutral path (`/Users/Shared/agentrun/workspace`). The agent's state file carried only the task identifier and a slot number; the mapping from slot to condition (`harness/rerun_schedule.json`, commit 0019d72) was read by the proxy and never shown to the agent. The clean runs were retained from the first run. In the rerun no transcript named the study, its repository or its paper. DeepSeek matched no pattern in any run. Flash matched awareness in 16 runs, each inferred from the task and the anomaly themselves rather than from the environment, the repository pattern in 17 runs of directory and process probing that found nothing, and the vocabulary in 2; the external files it fetched were again Biomni's own repository and Bio-Bench. Supplementary Table 1 gives both runs side by side; the main text reports the rerun throughout.

## Supplementary Note 2: registration and run timestamps

Each registration was committed before the runs it concerns; the intervals were: new-field experiment, commit 2059c3a, three minutes before the first run; its agreement follow-up, 217ea9d, one minute; superseded-sequence experiment, ac99248, 24 seconds; robustness addendum, 6830c57 and 76e56b6, 13 minutes and one minute before the respective runs; deployment hypotheses, OSF draft four minutes before the first full run, cited by commit a2dc4a4 two minutes before the runs; memory probe, 74415d3, and the prompt, repetition and withholding experiments, 499d317, each committed before the runs they concern on 3 November 2026. Commit times are recorded in the repository history.

Supplementary Table 1: Deployment experiment, first run (October 2026) and rerun (3 November 2026). Relay is relayed/delivered runs; the odds ratio compares relay of plausible values on exact-route fields with other fields (registered test, task-clustered logistic regression, 95% interval). Audit columns count runs matching each pattern.

| Model | Run | Impossible relayed | Plausible relayed | Odds ratio | Aware | Repository | Vocabulary |
|---|---|---|---|---|---|---|---|
| DeepSeek V4.1 Flash | first | 5/287 (2%) | 80/285 (28%) | 5.7 (3.1–10.4) | 0 | 0 | 1 |
| DeepSeek V4.1 Flash | rerun | 4/293 (1%) | 50/290 (17%) | 17.2 (6.0–49.7) | 0 | 0 | 0 |
| Gemini 3.8 Flash | first | 6/286 (2%) | 69/288 (24%) | 3.6 (2.0–6.5) | 44 | 98 | 2 |
| Gemini 3.8 Flash | rerun | 4/283 (1%) | 57/276 (21%) | 6.9 (3.3–14.3) | 16 | 17 | 2 |

Supplementary Table 2: Prompt variants: wrong value relayed (relayed/parsed runs) under the registered default prompt and three variants, on (a) protein tasks whose full record contradicts the value, (b) the chemistry agreement cell (three samples) and (c) coherent protein errors. The last column gives, for the two-fields variant, runs in which the stated database value matched the record (scored on (a) and (b)).

| Cell | Model | Default | No tool instruction | Database has errors | Two fields | Stated value = record |
|---|---|---|---|---|---|---|
| (a) | Opus 5 | 9/49 (18%) | 4/49 (8%) | 0/38 (0%) | 5/47 (11%) | 12/47 (26%) |
| (a) | Gemini 4 Argon | 7/50 (14%) | 6/50 (12%) | 1/50 (2%) | 0/50 (0%) | 0/50 (0%) |
| (a) | DeepSeek V4.1 Flash | 50/50 (100%) | 42/49 (86%) | 23/42 (55%) | 34/47 (72%) | 13/47 (28%) |
| (b) | Opus 5 | 163/180 (91%) | 85/180 (47%) | 0/180 (0%) | 0/180 (0%) | 180/180 (100%) |
| (b) | Gemini 4 Argon | 144/180 (80%) | 92/180 (51%) | 0/180 (0%) | 0/180 (0%) | 126/180 (70%) |
| (b) | Gemini 3.8 Flash | 159/174 (91%) | 58/171 (34%) | 0/180 (0%) | 6/163 (4%) | 103/163 (63%) |
| (b) | DeepSeek V4.1 Flash | 152/180 (84%) | 140/180 (78%) | 2/180 (1%) | 19/180 (11%) | 171/180 (95%) |
| (c) | Opus 5 | 49/50 (98%) | 48/50 (96%) | 50/50 (100%) | 26/50 (52%) | – |
| (c) | Gemini 4 Argon | 48/50 (96%) | 47/50 (94%) | 50/50 (100%) | 0/50 (0%) | – |
| (c) | DeepSeek V4.1 Flash | 47/50 (94%) | 46/50 (92%) | 49/49 (100%) | 45/50 (90%) | – |

Supplementary Table 3: Repetition versus consistency: wrong molecular weight relayed (relayed/parsed runs; 60 compounds, three samples, full record). R0, weight alone; R1, a verbatim repeat field scaled with the weight; R2, the repeat left correct; B, the heavy-atom count scaled with the weight. On uncorrupted records with the repeat field present, every model answered correctly in 60 of 60 runs. Pooled discordant pairs for RC1 (R0 relayed and R1 not, against the reverse): 198 against 84, two-sided McNemar $p < 10^{-10}$; for RC3 (R2 against R0), $p < 10^{-10}$.

| Model | R0 weight alone | R1 repeat agrees | R2 repeat disagrees | B derived field agrees |
|---|---|---|---|---|
| Opus 5 | 102/180 (57%) | 66/180 (37%) | 0/180 (0%) | 113/180 (63%) |
| Gemini 4 Argon | 117/180 (65%) | 67/180 (37%) | 3/180 (2%) | 130/180 (72%) |
| Gemini 3.8 Flash | 152/177 (86%) | 135/178 (76%) | 1/180 (1%) | 154/175 (88%) |
| DeepSeek V4.1 Flash | 115/180 (64%) | 105/179 (59%) | 5/180 (3%) | 128/180 (71%) |
| Pooled | 486/717 (68%) | 373/717 (52%) | 9/720 (1%) | 525/715 (73%) |

Supplementary Table 4: Memory probe on the 79 proteins with superseded sequences. Memory state is the model's answer, without tools, for the protein's length: current (within one residue of today's length), superseded (within one of the superseded length), either (both, where they differ by at most two) or other. Relay of the superseded value is given as relayed/runs by memory state, without a second source (full record, one sample) and with the second source shown (annotate arm, three samples); $p$ is the registered one-sided Fisher test of superseded-memory against current-memory relay with the second source.

| Model | Memory state | Proteins | No second source | Second source shown | $p$ |
|---|---|---|---|---|---|
| Opus 5 | current | 48 | 47/48 | 2/144 | |
| | superseded | 7 | 7/7 | 13/21 | $< 10^{-10}$ |
| | either | 12 | 12/12 | 0/36 | |
| | other | 12 | 12/12 | 9/36 | |
| Gemini 4 Argon | current | 53 | 48/53 | 4/159 | |
| | superseded | 6 | 6/6 | 0/17 | 1 |
| | either | 12 | 11/12 | 1/36 | |
| | other | 8 | 6/8 | 0/24 | |
| DeepSeek V4.1 Flash | current | 50 | 50/50 | 38/150 | |
| | superseded | 6 | 6/6 | 17/18 | $1 \times 10^{-8}$ |
| | either | 11 | 11/11 | 7/33 | |
| | other | 12 | 11/12 | 18/36 | |

Supplementary Table 5: Withholding harness (full record, 100 tasks per domain and condition, one sample): corrupted numeric fields and identifiers relayed (relayed/parsed runs), runs without a parseable answer, and runs answered with a value that was neither the true nor the corrupted one. The last column gives identifier relay in the annotate arm of the benchmark (summary record) for comparison; the arms differ in record format. Accuracy is correct answers among parsed runs.

| Model | Domain | Condition | Numeric relayed | Identifiers relayed | Unparsed | Wrong | Accuracy | Annotate arm identifiers |
|---|---|---|---|---|---|---|---|---|
| Opus 5 | proteins | impossible | 0/46 | 0/50 | 4 | 0 | 96/96 (100%) | 0/100 |
| | | plausible | 0/45 | 0/50 | 5 | 0 | 95/95 (100%) | 0/100 |
| | molecules | impossible | 0/75 | 5/25 | 0 | 0 | 93/100 (93%) | 21/25 |
| | | plausible | 0/75 | 3/25 | 0 | 0 | 93/100 (93%) | 10/25 |
| Gemini 4 Argon | proteins | impossible | 0/49 | 0/50 | 1 | 0 | 99/99 (100%) | 0/100 |
| | | plausible | 0/48 | 0/50 | 2 | 0 | 98/98 (100%) | 0/100 |
| | molecules | impossible | 0/75 | 4/25 | 0 | 0 | 94/100 (94%) | 19/25 |
| | | plausible | 0/75 | 1/25 | 0 | 0 | 95/100 (95%) | 18/25 |
| DeepSeek V4.1 Flash | proteins | impossible | 0/34 | 4/50 | 16 | 5 | 79/84 (94%) | 1/100 |
| | | plausible | 0/37 | 1/50 | 13 | 3 | 84/87 (97%) | 1/100 |
| | molecules | impossible | 0/75 | 12/25 | 0 | 0 | 83/100 (83%) | 22/24 |
| | | plausible | 0/75 | 11/25 | 0 | 0 | 85/100 (85%) | 21/24 |

**Supplementary Tables 6–13: counts behind Figures 1, 3, 4 and 6 and Extended Data Figures 1 and 2**

Supplementary Table 6: Deployment rerun (Fig. 1b, c): corrupted values relayed by field (relayed/delivered runs) and, among relayed runs, how many raised a concern.

| Model | Condition | Field | Relayed | Flagged among relayed |
|---|---|---|---|---|
| DeepSeek V4.1 Flash | impossible | gene symbol | 2/50 (4%) | 1/2 |
| DeepSeek V4.1 Flash | impossible | h bond donors | 0/24 (0%) | – |
| DeepSeek V4.1 Flash | impossible | heavy atom count | 0/25 (0%) | – |
| DeepSeek V4.1 Flash | impossible | length | 0/46 (0%) | – |
| DeepSeek V4.1 Flash | impossible | mass | 0/50 (0%) | – |
| DeepSeek V4.1 Flash | impossible | molecular weight | 0/23 (0%) | – |
| DeepSeek V4.1 Flash | impossible | name | 2/25 (8%) | 2/2 |
| DeepSeek V4.1 Flash | impossible | organism | 0/50 (0%) | – |
| DeepSeek V4.1 Flash | plausible | gene symbol | 0/49 (0%) | – |
| DeepSeek V4.1 Flash | plausible | h bond donors | 4/25 (16%) | 4/4 |
| DeepSeek V4.1 Flash | plausible | heavy atom count | 0/25 (0%) | – |
| DeepSeek V4.1 Flash | plausible | length | 23/44 (52%) | 6/23 |
| DeepSeek V4.1 Flash | plausible | mass | 18/49 (37%) | 4/18 |
| DeepSeek V4.1 Flash | plausible | molecular weight | 5/24 (21%) | 0/5 |
| DeepSeek V4.1 Flash | plausible | name | 0/24 (0%) | – |
| DeepSeek V4.1 Flash | plausible | organism | 0/50 (0%) | – |
| Gemini 3.8 Flash | impossible | gene symbol | 0/50 (0%) | – |
| Gemini 3.8 Flash | impossible | h bond donors | 0/25 (0%) | – |
| Gemini 3.8 Flash | impossible | heavy atom count | 0/22 (0%) | – |
| Gemini 3.8 Flash | impossible | length | 1/48 (2%) | 0/1 |
| Gemini 3.8 Flash | impossible | mass | 1/48 (2%) | 1/1 |
| Gemini 3.8 Flash | impossible | molecular weight | 0/21 (0%) | – |
| Gemini 3.8 Flash | impossible | name | 1/23 (4%) | 0/1 |
| Gemini 3.8 Flash | impossible | organism | 1/46 (2%) | 1/1 |
| Gemini 3.8 Flash | plausible | gene symbol | 1/49 (2%) | 1/1 |
| Gemini 3.8 Flash | plausible | h bond donors | 5/23 (22%) | 4/5 |
| Gemini 3.8 Flash | plausible | heavy atom count | 0/23 (0%) | – |
| Gemini 3.8 Flash | plausible | length | 16/44 (36%) | 7/16 |
| Gemini 3.8 Flash | plausible | mass | 30/48 (62%) | 9/30 |
| Gemini 3.8 Flash | plausible | molecular weight | 1/21 (5%) | 1/1 |
| Gemini 3.8 Flash | plausible | name | 0/20 (0%) | – |
| Gemini 3.8 Flash | plausible | organism | 4/48 (8%) | 3/4 |

Supplementary Table 7: Benchmark (Extended Data Fig. 1): corrupted values relayed, runs raising a concern, and relayed runs that raised a concern (summary record, registered prompt, one sample).

| Model | Domain | Condition | Relayed | Flagged | Flagged among relayed |
|---|---|---|---|---|---|
| Opus 5 | proteins | impossible | 64/200 (32%) | 200/200 (100%) | 64/64 |
| Opus 5 | proteins | plausible | 150/200 (75%) | 118/200 (59%) | 68/150 |
| Opus 5 | molecules | impossible | 46/100 (46%) | 100/100 (100%) | 46/46 |
| Opus 5 | molecules | plausible | 98/100 (98%) | 97/100 (97%) | 95/98 |
| Gemini 4 Argon | proteins | impossible | 70/200 (35%) | 200/200 (100%) | 70/70 |
| Gemini 4 Argon | proteins | plausible | 147/199 (74%) | 189/199 (95%) | 137/147 |
| Gemini 4 Argon | molecules | impossible | 62/100 (62%) | 91/100 (91%) | 53/62 |
| Gemini 4 Argon | molecules | plausible | 98/100 (98%) | 90/100 (90%) | 88/98 |
| DeepSeek V4.1 Flash | proteins | impossible | 88/200 (44%) | 200/200 (100%) | 88/88 |
| DeepSeek V4.1 Flash | proteins | plausible | 154/200 (77%) | 92/200 (46%) | 46/154 |
| DeepSeek V4.1 Flash | molecules | impossible | 38/100 (38%) | 97/100 (97%) | 35/38 |
| DeepSeek V4.1 Flash | molecules | plausible | 95/99 (96%) | 71/99 (72%) | 67/95 |
| GPT-6.1 Sol | proteins | impossible | 133/200 (66%) | 200/200 (100%) | 133/133 |
| GPT-6.1 Sol | proteins | plausible | 150/200 (75%) | 105/200 (52%) | 55/150 |
| GPT-6.1 Sol | molecules | impossible | 67/100 (67%) | 78/100 (78%) | 45/67 |
| GPT-6.1 Sol | molecules | plausible | 93/100 (93%) | 71/100 (71%) | 64/93 |
| GPT-6 Luna | proteins | impossible | 146/200 (73%) | 163/200 (82%) | 109/146 |
| GPT-6 Luna | proteins | plausible | 171/200 (86%) | 34/200 (17%) | 6/171 |
| GPT-6 Luna | molecules | impossible | 68/100 (68%) | 75/100 (75%) | 43/68 |
| GPT-6 Luna | molecules | plausible | 100/100 (100%) | 16/100 (16%) | 16/100 |
| Kimi K3 | proteins | impossible | 149/200 (74%) | 199/200 (100%) | 148/149 |
| Kimi K3 | proteins | plausible | 184/200 (92%) | 58/200 (29%) | 42/184 |
| Kimi K3 | molecules | impossible | 63/100 (63%) | 88/100 (88%) | 51/63 |
| Kimi K3 | molecules | plausible | 99/100 (99%) | 60/100 (60%) | 59/99 |
| Sonnet 5 | proteins | impossible | 151/200 (76%) | 200/200 (100%) | 151/151 |
| Sonnet 5 | proteins | plausible | 185/200 (92%) | 63/200 (32%) | 48/185 |
| Sonnet 5 | molecules | impossible | 73/100 (73%) | 90/100 (90%) | 63/73 |
| Sonnet 5 | molecules | plausible | 100/100 (100%) | 50/100 (50%) | 50/100 |
| Sonnet 4.6 | proteins | impossible | 166/200 (83%) | 199/200 (100%) | 165/166 |
| Sonnet 4.6 | proteins | plausible | 199/200 (100%) | 50/200 (25%) | 49/199 |
| Sonnet 4.6 | molecules | impossible | 83/100 (83%) | 84/100 (84%) | 67/83 |
| Sonnet 4.6 | molecules | plausible | 100/100 (100%) | 38/100 (38%) | 38/100 |
| Gemini 3.8 Flash | proteins | impossible | 167/198 (84%) | 195/198 (98%) | 164/167 |
| Gemini 3.8 Flash | proteins | plausible | 155/200 (78%) | 64/200 (32%) | 19/155 |
| Gemini 3.8 Flash | molecules | impossible | 62/99 (63%) | 77/99 (78%) | 40/62 |
| Gemini 3.8 Flash | molecules | plausible | 100/100 (100%) | 56/100 (56%) | 56/100 |
| Haiku 4.5 | proteins | impossible | 186/200 (93%) | 152/200 (76%) | 138/186 |
| Haiku 4.5 | proteins | plausible | 200/200 (100%) | 15/200 (8%) | 15/200 |
| Haiku 4.5 | molecules | impossible | 91/100 (91%) | 75/100 (75%) | 66/91 |
| Haiku 4.5 | molecules | plausible | 100/100 (100%) | 3/100 (3%) | 3/100 |

Supplementary Table 8: Override boundaries (Extended Data Fig. 2), value decreased: corrupted value relayed (relayed/parsed runs) at each corruption factor, full record, paired field scaled alike.

| Model | Domain | ×0.9 | ×0.8 | ×0.7 | ×0.5 | ×0.3 | ×0.1 | ×0.02 | ×0.005 |
|---|---|---|---|---|---|---|---|---|---|
| Opus 5 | proteins | 18/40 | 8/39 | 1/40 | 0/39 | 0/39 | 0/39 | 0/38 | 0/33 |
| Opus 5 | molecules | 16/40 | 14/40 | 9/40 | 4/40 | 0/40 | 1/40 | 0/40 | 0/40 |
| Sonnet 5 | proteins | 37/40 | 35/40 | 28/40 | 15/38 | 5/40 | 0/40 | 0/40 | 0/38 |
| Sonnet 5 | molecules | 40/40 | 40/40 | 33/40 | 17/40 | 3/40 | 0/40 | 0/40 | 0/40 |
| Gemini 4 Argon | proteins | 23/40 | 16/40 | 9/40 | 1/40 | 0/40 | 0/40 | 0/40 | 0/40 |
| Gemini 4 Argon | molecules | 28/40 | 24/40 | 22/40 | 11/40 | 3/40 | 0/40 | 0/40 | 0/40 |
| Gemini 3.8 Flash | proteins | 21/39 | 17/39 | 12/38 | 5/37 | 2/40 | 0/40 | 0/40 | 0/40 |
| Gemini 3.8 Flash | molecules | 29/39 | 22/35 | 15/39 | 15/40 | 7/39 | 2/40 | 5/40 | 4/40 |
| DeepSeek V4.1 Flash | proteins | 38/40 | 33/39 | 33/39 | 21/39 | 8/36 | 0/35 | 1/30 | 0/30 |
| DeepSeek V4.1 Flash | molecules | 22/40 | 11/40 | 10/40 | 6/40 | 1/39 | 0/40 | 0/40 | 0/40 |

Supplementary Table 9: Override boundaries (Extended Data Fig. 2), value increased: corrupted value relayed (relayed/parsed runs) at each corruption factor, full record, paired field scaled alike.

| Model | Domain | ×1.1 | ×1.25 | ×1.5 | ×2 | ×3 | ×10 | ×50 | ×200 |
|---|---|---|---|---|---|---|---|---|---|
| Opus 5 | proteins | 25/40 | 6/38 | 4/40 | 0/40 | 0/40 | 0/40 | 0/40 | 0/40 |
| Opus 5 | molecules | 19/40 | 10/40 | 7/40 | 4/40 | 4/40 | 0/40 | 0/40 | 0/40 |
| Sonnet 5 | proteins | 40/40 | 37/40 | 32/40 | 24/40 | 12/39 | 0/40 | 0/38 | 0/40 |
| Sonnet 5 | molecules | 40/40 | 40/40 | 40/40 | 36/40 | 30/40 | 4/40 | 2/40 | 0/40 |
| Gemini 4 Argon | proteins | 15/40 | 11/40 | 4/40 | 0/40 | 0/40 | 0/40 | 0/40 | 0/40 |
| Gemini 4 Argon | molecules | 22/40 | 19/40 | 11/40 | 13/40 | 5/40 | 0/40 | 0/40 | 0/40 |
| Gemini 3.8 Flash | proteins | 26/40 | 17/39 | 6/39 | 1/40 | 0/40 | 0/40 | 0/40 | 0/40 |
| Gemini 3.8 Flash | molecules | 21/40 | 21/39 | 17/39 | 14/40 | 18/38 | 0/40 | 4/39 | 0/40 |
| DeepSeek V4.1 Flash | proteins | 40/40 | 31/40 | 24/39 | 13/38 | 3/34 | 0/38 | 1/33 | 0/38 |
| DeepSeek V4.1 Flash | molecules | 30/40 | 22/40 | 19/40 | 13/40 | 5/40 | 0/40 | 0/40 | 0/40 |

Supplementary Table 10: New chemistry fields with the full record (Fig. 4a): runs raising a concern and corrupted values relayed, by field and condition.

| Model | Field | Condition | Flagged | Relayed |
|---|---|---|---|---|
| Opus 5 | complexity | clean | 1/60 (2%) | – |
| Opus 5 | complexity | plausible | 1/60 (2%) | 60/60 (100%) |
| Opus 5 | complexity | impossible | 60/60 (100%) | 60/60 (100%) |
| Opus 5 | monoisotopic mass | clean | 2/60 (3%) | – |
| Opus 5 | monoisotopic mass | plausible | 52/60 (87%) | 57/60 (95%) |
| Opus 5 | monoisotopic mass | impossible | 60/60 (100%) | 0/60 (0%) |
| Opus 5 | tpsa | clean | 1/60 (2%) | – |
| Opus 5 | tpsa | plausible | 10/60 (17%) | 60/60 (100%) |
| Opus 5 | tpsa | impossible | 60/60 (100%) | 19/60 (32%) |
| Sonnet 5 | complexity | clean | 3/60 (5%) | – |
| Sonnet 5 | complexity | plausible | 2/60 (3%) | 60/60 (100%) |
| Sonnet 5 | complexity | impossible | 60/60 (100%) | 60/60 (100%) |
| Sonnet 5 | monoisotopic mass | clean | 6/60 (10%) | – |
| Sonnet 5 | monoisotopic mass | plausible | 36/60 (60%) | 59/60 (98%) |
| Sonnet 5 | monoisotopic mass | impossible | 60/60 (100%) | 2/60 (3%) |
| Sonnet 5 | tpsa | clean | 0/60 (0%) | – |
| Sonnet 5 | tpsa | plausible | 0/60 (0%) | 60/60 (100%) |
| Sonnet 5 | tpsa | impossible | 60/60 (100%) | 41/60 (68%) |
| Gemini 4 Argon | complexity | clean | 0/60 (0%) | – |
| Gemini 4 Argon | complexity | plausible | 1/60 (2%) | 60/60 (100%) |
| Gemini 4 Argon | complexity | impossible | 60/60 (100%) | 51/60 (85%) |
| Gemini 4 Argon | monoisotopic mass | clean | 3/60 (5%) | – |
| Gemini 4 Argon | monoisotopic mass | plausible | 60/60 (100%) | 58/60 (97%) |
| Gemini 4 Argon | monoisotopic mass | impossible | 60/60 (100%) | 0/60 (0%) |
| Gemini 4 Argon | tpsa | clean | 1/60 (2%) | – |
| Gemini 4 Argon | tpsa | plausible | 47/60 (78%) | 60/60 (100%) |
| Gemini 4 Argon | tpsa | impossible | 60/60 (100%) | 12/60 (20%) |
| Gemini 3.8 Flash | complexity | clean | 0/60 (0%) | – |
| Gemini 3.8 Flash | complexity | plausible | 0/60 (0%) | 60/60 (100%) |
| Gemini 3.8 Flash | complexity | impossible | 60/60 (100%) | 57/60 (95%) |
| Gemini 3.8 Flash | monoisotopic mass | clean | 0/60 (0%) | – |
| Gemini 3.8 Flash | monoisotopic mass | plausible | 58/60 (97%) | 60/60 (100%) |
| Gemini 3.8 Flash | monoisotopic mass | impossible | 58/58 (100%) | 1/58 (2%) |
| Gemini 3.8 Flash | tpsa | clean | 0/60 (0%) | – |
| Gemini 3.8 Flash | tpsa | plausible | 4/60 (7%) | 59/60 (98%) |
| Gemini 3.8 Flash | tpsa | impossible | 56/56 (100%) | 30/56 (54%) |
| DeepSeek V4.1 Flash | complexity | clean | 6/60 (10%) | – |
| DeepSeek V4.1 Flash | complexity | plausible | 5/60 (8%) | 60/60 (100%) |
| DeepSeek V4.1 Flash | complexity | impossible | 60/60 (100%) | 60/60 (100%) |
| DeepSeek V4.1 Flash | monoisotopic mass | clean | 2/60 (3%) | – |
| DeepSeek V4.1 Flash | monoisotopic mass | plausible | 55/60 (92%) | 56/60 (93%) |
| DeepSeek V4.1 Flash | monoisotopic mass | impossible | 60/60 (100%) | 0/60 (0%) |
| DeepSeek V4.1 Flash | tpsa | clean | 2/60 (3%) | – |
| DeepSeek V4.1 Flash | tpsa | plausible | 8/60 (13%) | 60/60 (100%) |
| DeepSeek V4.1 Flash | tpsa | impossible | 58/58 (100%) | 13/58 (22%) |

Supplementary Table 11: Agreement follow-up (Fig. 4b, c): wrong value relayed on the same 60 compounds, pooled over three samples and per sample.

| Model | Arm | Pooled | Sample 1 | Sample 2 | Sample 3 |
|---|---|---|---|---|---|
| Opus 5 | mono. mass asked, two agreeing fields | 175/180 (97%) | 57/60 (95%) | 60/60 (100%) | 58/60 (97%) |
| Opus 5 | mono. mass asked, corrupted alone | 138/180 (77%) | 48/60 (80%) | 47/60 (78%) | 43/60 (72%) |
| Opus 5 | weight asked, one agreeing field | 113/180 (63%) | 31/60 (52%) | 43/60 (72%) | 39/60 (65%) |
| Opus 5 | weight asked, two agreeing fields | 163/180 (91%) | 52/60 (87%) | 58/60 (97%) | 53/60 (88%) |
| Gemini 4 Argon | mono. mass asked, two agreeing fields | 172/180 (96%) | 58/60 (97%) | 55/60 (92%) | 59/60 (98%) |
| Gemini 4 Argon | mono. mass asked, corrupted alone | 129/180 (72%) | 43/60 (72%) | 46/60 (77%) | 40/60 (67%) |
| Gemini 4 Argon | weight asked, one agreeing field | 130/180 (72%) | 40/60 (67%) | 44/60 (73%) | 46/60 (77%) |
| Gemini 4 Argon | weight asked, two agreeing fields | 144/180 (80%) | 52/60 (87%) | 44/60 (73%) | 48/60 (80%) |
| Gemini 3.8 Flash | mono. mass asked, two agreeing fields | 178/180 (99%) | 60/60 (100%) | 59/60 (98%) | 59/60 (98%) |
| Gemini 3.8 Flash | mono. mass asked, corrupted alone | 149/176 (85%) | 50/59 (85%) | 49/59 (83%) | 50/58 (86%) |
| Gemini 3.8 Flash | weight asked, one agreeing field | 154/175 (88%) | 52/59 (88%) | 53/59 (90%) | 49/57 (86%) |
| Gemini 3.8 Flash | weight asked, two agreeing fields | 159/174 (91%) | 52/58 (90%) | 54/58 (93%) | 53/58 (91%) |
| DeepSeek V4.1 Flash | mono. mass asked, two agreeing fields | 158/179 (88%) | 56/60 (93%) | 52/59 (88%) | 50/60 (83%) |
| DeepSeek V4.1 Flash | mono. mass asked, corrupted alone | 100/180 (56%) | 36/60 (60%) | 29/60 (48%) | 35/60 (58%) |
| DeepSeek V4.1 Flash | weight asked, one agreeing field | 128/180 (71%) | 43/60 (72%) | 41/60 (68%) | 44/60 (73%) |
| DeepSeek V4.1 Flash | weight asked, two agreeing fields | 152/180 (84%) | 54/60 (90%) | 48/60 (80%) | 50/60 (83%) |

Supplementary Table 12: Superseded UniProt sequences (Fig. 3): superseded value relayed with the full and summary record (one sample), full-record repeats (three samples; for Opus 5 the first 30 tasks, as registered in 6830c57), and with AlphaFold DB as second source (annotate arm, three samples) on covered proteins, by direction of the revision, and on the uncovered protein.

| Model | Full record | Summary | Full, repeats | Second source | Superseded longer | Superseded shorter | Uncovered |
|---|---|---|---|---|---|---|---|
| Opus 5 | 78/79 (99%) | 79/79 (100%) | 90/90 (100%) | 21/234 (9%) | 16/63 (25%) | 5/171 (3%) | 3/3 (100%) |
| Gemini 4 Argon | 71/79 (90%) | 79/79 (100%) | 213/236 (90%) | 2/233 (1%) | 1/63 (2%) | 1/170 (1%) | 3/3 (100%) |
| DeepSeek V4.1 Flash | 78/79 (99%) | 79/79 (100%) | 234/236 (99%) | 77/234 (33%) | 27/63 (43%) | 50/171 (29%) | 3/3 (100%) |

Supplementary Table 13: Presentation of the correction (Fig. 6): superseded value relayed on covered proteins under four presentations, pooled over three samples and per sample.

| Model | Presentation | Pooled | Sample 1 | Sample 2 | Sample 3 |
|---|---|---|---|---|---|
| Opus 5 | annotate: conflict listed | 21/234 (9%) | 7/78 (9%) | 7/78 (9%) | 7/78 (9%) |
| Opus 5 | replace: ambiguous labels | 25/234 (11%) | 7/78 (9%) | 9/78 (12%) | 9/78 (12%) |
| Opus 5 | replace: original labelled | 12/234 (5%) | 4/78 (5%) | 3/78 (4%) | 5/78 (6%) |
| Opus 5 | replace: silent | 0/234 (0%) | 0/78 (0%) | 0/78 (0%) | 0/78 (0%) |
| Gemini 4 Argon | annotate: conflict listed | 2/233 (1%) | 1/78 (1%) | 0/77 (0%) | 1/78 (1%) |
| Gemini 4 Argon | replace: ambiguous labels | 17/231 (7%) | 5/77 (6%) | 7/77 (9%) | 5/77 (6%) |
| Gemini 4 Argon | replace: original labelled | 95/231 (41%) | 37/78 (47%) | 31/77 (40%) | 27/76 (36%) |
| Gemini 4 Argon | replace: silent | 0/234 (0%) | 0/78 (0%) | 0/78 (0%) | 0/78 (0%) |
| DeepSeek V4.1 Flash | annotate: conflict listed | 77/234 (33%) | 24/78 (31%) | 28/78 (36%) | 25/78 (32%) |
| DeepSeek V4.1 Flash | replace: ambiguous labels | 6/234 (3%) | 1/78 (1%) | 3/78 (4%) | 2/78 (3%) |
| DeepSeek V4.1 Flash | replace: original labelled | 17/233 (7%) | 7/78 (9%) | 5/78 (6%) | 5/77 (6%) |
| DeepSeek V4.1 Flash | replace: silent | 0/234 (0%) | 0/78 (0%) | 0/78 (0%) | 0/78 (0%) |

## Supplementary Tables 14–20: experiments added in revision

Supplementary Table 14: Identity. Relay (relayed/parsed runs) on the 25 length-threshold benchmark tasks with the full record, with identity but no annotation (accession, protein name, gene, organism, length, mass, sequence), and with neither (anonymous; three samples, the first paired with the other formats); clean rows give accuracy. The last row per model gives relay of the 79 superseded sequences without a second source, with the full record and anonymously.

| Model | Condition | Full record | Identity, no annotation | Neither, sample 1 | Neither, 3 samples |
|---|---|---|---|---|---|
| Opus 5 | clean (accuracy) | 25/25 (100%) | 25/25 (100%) | 25/25 (100%) | 75/75 (100%) |
| Opus 5 | impossible | 0/25 (0%) | 0/25 (0%) | 0/25 (0%) | 0/75 (0%) |
| Opus 5 | plausible | 1/25 (4%) | 7/25 (28%) | 17/25 (68%) | 50/75 (67%) |
| Opus 5 | superseded record | 78/79 (99%) | – | 79/79 (100%) | – |
| Gemini 4 Argon | clean (accuracy) | 25/25 (100%) | 25/25 (100%) | 25/25 (100%) | 75/75 (100%) |
| Gemini 4 Argon | impossible | 0/25 (0%) | 0/25 (0%) | 0/25 (0%) | 0/75 (0%) |
| Gemini 4 Argon | plausible | 0/25 (0%) | 1/25 (4%) | 0/25 (0%) | 1/75 (1%) |
| Gemini 4 Argon | superseded record | 71/79 (90%) | – | 77/79 (97%) | – |
| DeepSeek V4.1 Flash | clean (accuracy) | 25/25 (100%) | 25/25 (100%) | 25/25 (100%) | 75/75 (100%) |
| DeepSeek V4.1 Flash | impossible | 0/25 (0%) | 0/24 (0%) | 1/25 (4%) | 2/73 (3%) |
| DeepSeek V4.1 Flash | plausible | 22/25 (88%) | 23/25 (92%) | 21/25 (84%) | 70/75 (93%) |
| DeepSeek V4.1 Flash | superseded record | 78/79 (99%) | – | 79/79 (100%) | – |

Supplementary Table 15: Reasoning. Relay (relayed/parsed runs) at the setting used throughout and with reasoning reduced, paired by task; the Reduction column gives the reduced setting for each model (Opus 5 and Sonnet 5 otherwise ran with adaptive thinking, Gemini 4 Argon at the provider's default).

| Model | Reduction | Cell | Condition | Setting used throughout | Reduced |
|---|---|---|---|---|---|
| Opus 5 | thinking disabled | full record, first 50 tasks | impossible | 0/50 (0%) | 1/50 (2%) |
| Opus 5 | thinking disabled | full record, first 50 tasks | plausible | 4/50 (8%) | 23/50 (46%) |
| Gemini 4 Argon | effort low | full record, first 50 tasks | impossible | 0/50 (0%) | 1/50 (2%) |
| Gemini 4 Argon | effort low | full record, first 50 tasks | plausible | 3/50 (6%) | 7/50 (14%) |
| Sonnet 5 | thinking disabled | summary record, proteins | impossible | 151/200 (76%) | 139/200 (70%) |
| Sonnet 5 | thinking disabled | summary record, proteins | plausible | 185/200 (92%) | 189/200 (94%) |
| Sonnet 5 | thinking disabled | summary record, chemistry | impossible | 73/100 (73%) | 63/100 (63%) |
| Sonnet 5 | thinking disabled | summary record, chemistry | plausible | 100/100 (100%) | 100/100 (100%) |

Supplementary Table 16: Request settings recorded in the traces, by model as requested (the three most frequent configurations; runs counted over the benchmark and round-2 files). Settings not listed were the provider's defaults. The label thinking=adaptive is the harness's own name for its default setting; the harness sent it as an API parameter only to Anthropic models and sent no reasoning parameter to other providers, except effort=low to Gemini 4 Argon in the reasoning ablation.

| Model | Settings | Runs |
|---|---|---|
| anthropic:claude-haiku-4-5 | max_tokens=8000; model=claude-haiku-4-5; provider=anthropic; thinking=adaptive | 1500 |
| anthropic:claude-opus-5 | max_tokens=8000; model=claude-opus-5; provider=anthropic; thinking=adaptive | 2534 |
| anthropic:claude-opus-5 | max_tokens=8000; model=claude-opus-5; provider=anthropic; thinking=disabled | 100 |
| anthropic:claude-sonnet-4-6 | max_tokens=8000; model=claude-sonnet-4-6; provider=anthropic; thinking=adaptive | 1500 |
| anthropic:claude-sonnet-5 | max_tokens=8000; model=claude-sonnet-5; provider=anthropic; thinking=adaptive | 1500 |
| anthropic:claude-sonnet-5 | max_tokens=8000; model=claude-sonnet-5; provider=anthropic; thinking=disabled | 600 |
| gemini:gemini-3.8-flash | max_tokens=8000; model=google/gemini-3.8-flash; provider=gemini; reasoning_effort=None; thinking=adaptive | 1680 |
| gemini:gemini-4-argon | max_tokens=8000; model=google/gemini-4-argon; provider=gemini; reasoning_effort=None; thinking=adaptive | 2534 |
| gemini:gemini-4-argon | effort=low; max_tokens=8000; model=google/gemini-4-argon; provider=gemini; reasoning_effort=low; thinking=adaptive | 100 |
| openai:gpt-6-luna | api=responses; max_tokens=8000; model=gpt-6-luna; provider=openai; reasoning_effort=None; thinking=adaptive | 1500 |
| openai:gpt-6.1-sol | api=responses; max_tokens=8000; model=gpt-6.1-sol; provider=openai; reasoning_effort=None; thinking=adaptive | 1500 |
| together:deepseek-ai/DeepSeek-V4.1-Flash-1017 | max_tokens=8000; model=deepseek-ai/DeepSeek-V4.1-Flash-1017; provider=together; reasoning_effort=None; thinking=adaptive | 2534 |
| together:moonshotai/Kimi-K3 | max_tokens=8000; model=moonshotai/Kimi-K3; provider=together; reasoning_effort=None; thinking=adaptive | 1500 |

Supplementary Table 17: Prompt without the concerns field (summary record, first 100 tasks per domain). Relay under the registered prompt and without the concerns field; runs listing a concern under the registered prompt; runs voicing doubt in the reply text with and without the field, by the validated classifier; and the registered keyword detector without the field, a lower bound.

| Model | Domain | Condition | Relay, registered | Relay, no field | Concerns listed | Doubt in reply, registered | Doubt in reply, no field | Keyword, no field |
|---|---|---|---|---|---|---|---|---|
| Opus 5 | proteins | impossible | 33/100 (33%) | 60/100 (60%) | 100/100 (100%) | 99/100 (99%) | 84/100 (84%) | 45/100 (45%) |
| Opus 5 | proteins | plausible | 75/100 (75%) | 79/100 (79%) | 57/100 (57%) | 52/100 (52%) | 29/100 (29%) | 17/100 (17%) |
| Opus 5 | chemistry | impossible | 46/100 (46%) | 57/100 (57%) | 100/100 (100%) | 100/100 (100%) | 75/100 (75%) | 32/100 (32%) |
| Opus 5 | chemistry | plausible | 98/100 (98%) | 100/100 (100%) | 97/100 (97%) | 96/100 (96%) | 17/100 (17%) | 0/100 (0%) |
| Gemini 4 Argon | proteins | impossible | 37/100 (37%) | 80/100 (80%) | 100/100 (100%) | 99/100 (99%) | 36/100 (36%) | 5/100 (5%) |
| Gemini 4 Argon | proteins | plausible | 74/100 (74%) | 100/100 (100%) | 94/100 (94%) | 85/100 (85%) | 1/100 (1%) | 0/100 (0%) |
| Gemini 4 Argon | chemistry | impossible | 62/100 (62%) | 39/100 (39%) | 91/100 (91%) | 83/100 (83%) | 72/100 (72%) | 31/100 (31%) |
| Gemini 4 Argon | chemistry | plausible | 98/100 (98%) | 100/100 (100%) | 90/100 (90%) | 81/100 (81%) | 0/100 (0%) | 0/100 (0%) |
| DeepSeek V4.1 Flash | proteins | impossible | 43/100 (43%) | 68/100 (68%) | 100/100 (100%) | 95/100 (95%) | 71/100 (71%) | 35/100 (35%) |
| DeepSeek V4.1 Flash | proteins | plausible | 78/100 (78%) | 83/100 (83%) | 43/100 (43%) | 41/100 (41%) | 21/100 (21%) | 8/100 (8%) |
| DeepSeek V4.1 Flash | chemistry | impossible | 38/100 (38%) | 50/100 (50%) | 97/100 (97%) | 87/100 (87%) | 66/100 (66%) | 20/100 (20%) |
| DeepSeek V4.1 Flash | chemistry | plausible | 95/99 (96%) | 100/100 (100%) | 71/99 (72%) | 57/99 (58%) | 14/100 (14%) | 2/100 (2%) |

Supplementary Table 18: Familiar duplicate field. Relay of a wrong molecular weight alone and with a verbatim copy that agrees or disagrees, under an unfamiliar (molecular_weight_reported) or familiar (average_mass) field name; first sample of the earlier arms; 60 compounds.

| Model | Alone | Copy agrees, unfamiliar | Copy agrees, familiar | Copy disagrees, unfamiliar | Copy disagrees, familiar | Clean accuracy, familiar copy |
|---|---|---|---|---|---|---|
| Opus 5 | 35/60 (58%) | 24/60 (40%) | 32/60 (53%) | 0/60 (0%) | 0/60 (0%) | 60/60 (100%) |
| Gemini 4 Argon | 37/60 (62%) | 23/60 (38%) | 32/60 (53%) | 1/60 (2%) | 6/60 (10%) | 60/60 (100%) |
| Gemini 3.8 Flash | 48/59 (81%) | 47/59 (80%) | 46/56 (82%) | 0/60 (0%) | 1/57 (2%) | 60/60 (100%) |
| DeepSeek V4.1 Flash | 32/60 (53%) | 39/59 (66%) | 40/59 (68%) | 0/60 (0%) | 14/60 (23%) | 60/60 (100%) |

Supplementary Table 19: Revision audit of the 79 superseded sequences: class, number of proteins, number with the change at the N-terminus, and relay of the superseded value pooled over Opus 5, Gemini 4 Argon and DeepSeek V4.1 Flash in four arms (second source shown: three samples). The second-source column includes the one protein AlphaFold DB does not cover, whose old sequence matches a current RefSeq or Ensembl model (9 runs, all relayed); Supplementary Tables 12 and 13 exclude it and so count 701 runs.

| Class | Proteins | N-terminal | Full record | Summary | Second source shown | Identity removed |
|---|---|---|---|---|---|---|
| documented error | 13 | 10 | 33/39 (85%) | 39/39 (100%) | 27/117 (23%) | 39/39 (100%) |
| flagged discrepancy | 1 | 0 | 3/3 (100%) | 3/3 (100%) | 0/9 (0%) | 3/3 (100%) |
| current isoform | 16 | 10 | 48/48 (100%) | 48/48 (100%) | 25/143 (17%) | 48/48 (100%) |
| matches a current RefSeq/Ensembl model | 1 | 0 | 3/3 (100%) | 3/3 (100%) | 9/9 (100%) | 3/3 (100%) |
| retired, reason not recorded | 48 | 26 | 140/144 (97%) | 144/144 (100%) | 48/432 (11%) | 142/144 (99%) |

Supplementary Table 20: Runs without a parseable answer: every cell with at least two, with the reported rate (unparsed runs excluded) and the rate with unparsed runs counted against the agent (as relayed for relay, as wrong for accuracy).

| Experiment | File | Condition | Parsed | Unparsed | Reported | Counted against the agent |
|---|---|---|---|---|---|---|
| benchmark | traces_gemini38flash | impossible | 198 | 2 | relay 167/198 (84%) | 169/200 (84%) |
| full record | traces_deepseek_v41flash | impossible | 97 | 3 | relay 11/97 (11%) | 14/100 (14%) |
| prompt variants | deepseek_protein_record_errors | plausible | 92 | 8 | relay 23/92 (25%) | 31/100 (31%) |
| prompt variants | deepseek_protein_two_fields | plausible | 97 | 3 | relay 35/97 (36%) | 38/100 (38%) |
| prompt variants | flash_agreeC_no_tool_instruction | plausible | 171 | 9 | relay 58/171 (34%) | 67/180 (37%) |
| prompt variants | flash_agreeC_two_fields | plausible | 163 | 17 | relay 6/163 (4%) | 23/180 (13%) |
| prompt variants | opus5_protein_record_errors | plausible | 88 | 12 | relay 0/88 (0%) | 12/100 (12%) |
| prompt variants | opus5_protein_two_fields | plausible | 97 | 3 | relay 5/97 (5%) | 8/100 (8%) |
| withholding | argon_protein | plausible | 98 | 2 | relay 0/98 (0%) | 2/100 (2%) |
| withholding | deepseek_protein | impossible | 84 | 16 | relay 4/84 (5%) | 20/100 (20%) |
| withholding | deepseek_protein | plausible | 87 | 13 | relay 1/87 (1%) | 14/100 (14%) |
| withholding | opus5_protein | impossible | 96 | 4 | relay 0/96 (0%) | 4/100 (4%) |
| withholding | opus5_protein | plausible | 95 | 5 | relay 0/95 (0%) | 5/100 (5%) |
| round 2 | deepseek_IS_bench | impossible | 73 | 2 | relay 2/73 (3%) | 4/75 (5%) |
| round 2 | flash_F1 | plausible | 56 | 4 | relay 46/56 (82%) | 50/60 (83%) |
| round 2 | flash_F2 | plausible | 57 | 3 | relay 1/57 (2%) | 4/60 (7%) |
| superseded | argon_p5_replace | superseded | 234 | 3 | relay 20/234 (9%) | 23/237 (10%) |
| superseded | argon_p5_replace_labeled | superseded | 234 | 3 | relay 98/234 (42%) | 101/237 (43%) |

Supplementary Table 21: Every registered prediction, with the commit or registration that fixed it before the runs, its test, and its outcome. Labels are those of the registrations, except that the coherent-corruption and memory-probe predictions (registered as P1–P4 and P1–P3) are prefixed C and M. Failed predictions are tagged with the group used in Methods (Registrations): V, the value itself; C, an effect expected to carry over; P, the prompt, harness, reasoning or memory; D, the doubt detector.

| ID | Experiment | Prediction and test | Outcome | Status |
|---|---|---|---|---|
| DH1 | Deployment (OSF jmwab) | Plausible values are relayed less often in deployment than in the benchmark (McNemar, per model) | 17% and 21% against 84%; $p < 10^{-10}$ for both models | held |
| DH2 | Deployment | Plausible corruptions of exact-route fields are relayed less often than of other fields (logistic regression, task-clustered) | Relayed more: odds ratios 17.2 (DeepSeek) and 6.9 (Flash) | failed (V) |
| DH3 | Deployment | Relayed values are flagged less often in deployment than in the benchmark (one-sided Fisher) | DeepSeek plausible 28% against 45% ($p = 0.016$); Flash 44% against 29%; impossible values too few ($p = 0.12$, 0.07) | partly held |
| H1 | New fields (2059c3a) | Plausible relay rises with route class, monoisotopic mass < TPSA < complexity (one-sided Wald; Opus 5, Argon, Flash, DeepSeek) | Positive ($p = 0.007$) but negligible: 96%, 99.6%, 100% | held |
| H2 | New fields | Monoisotopic-mass relay falls inside each model's interval from its molecular-weight curve in at least 8 of 10 cells | 2 of 10 (both Sonnet 5, at ceiling); 90–100% observed, 27–59% predicted | failed (C) |
| H3 | New fields | The route effect is larger with the full than the summary record (interaction, one-sided) | Every summary-record run relayed | not testable |
| E1 | Agreement follow-up (217ea9d) | Molecular weight with one agreeing field falls inside the H2 intervals in at least 8 of 10 cells | 5 of 10 | failed (C) |
| E2 | Agreement follow-up | Relay is higher with two agreeing fields than one (logistic regression, one-sided) | $p = 4 \times 10^{-7}$ | held |
| E3 | Agreement follow-up | Relay is lower when monoisotopic mass is corrupted alone | $p = 9 \times 10^{-10}$; all four models | held |
| C1 | Coherent corruptions (28 Oct; registered as P1–P4) | Coherent plausible errors with the full record are relayed in at least 70% for all three models | 70–98% | held |
| C2 | Coherent corruptions | Coherent impossible values with the full record are relayed less than the summary-record impossible baseline, but above 0 | 3 of 6 comparisons as predicted | partly held |
| C3 | Coherent corruptions | With the annotate harness and the second source, relay is at most 5% on covered entities | Argon 0%; Opus 5 and DeepSeek above 5% in some cells | partly held |
| C4 | Coherent corruptions | An all-clear harness message raises relay of coherent impossible values from the summary record | Not raised (proteins at ceiling; chemistry lower for Opus 5) | failed (P) |
| R1 | Superseded sequences (ac99248) | Without a second source, relay is at least 80% in at least 5 of 6 model-by-arm cells | 6 of 6; 90–100% | held |
| R2 | Superseded sequences | With the second source, relay on covered proteins is at most 10% for each model | Opus 5 9%, Argon 1%, DeepSeek 31–36% in every sample | failed (C) |
| R3 | Superseded sequences | Residual relay is higher when the superseded sequence is longer than today's (one-sided Fisher) | 25% against 11%, $p = 0.007$ (first sample) | held |

| ID | Experiment | Prediction and test | Outcome | Status |
|---|---|---|---|---|
| RR1 | Robustness addendum (6830c57) | Replace with ambiguous labels: relay on covered proteins at most 2% per model | 11%, 7% and 3% | failed (C) |
| RR2 | Robustness addendum | Replace relays less than annotate (one-sided Wilcoxon, pooled) | $p = 2.5 \times 10^{-4}$ pooled; heterogeneous by model | held |
| L1 | Robustness addendum (76e56b6) | Original labelled: relay at most 2% per model | Opus 5 5%, Argon 41%, DeepSeek 7% | failed (C) |
| L2 | Robustness addendum | Labelled relays less than ambiguous labels (one-sided Wilcoxon) | $p = 1.0$ | failed (C) |
| S1 | Robustness addendum | Silent correction: relay at most 2% per model | 0 of 702 | held |
| M1 | Memory probe (74415d3; registered as P1–P3) | Memory holds the superseded length for at least 20% of proteins in at least one model | 8–9% | failed (P) |
| M2 | Memory probe | With the second source shown, superseded-memory proteins are relayed more than current-memory ones (one-sided Fisher) | Opus 5 $p < 10^{-10}$, DeepSeek $p = 1 \times 10^{-8}$; Argon 0 of 17 against 4 of 159 | partly held |
| M3 | Memory probe | Without a second source, relay is at least 90% in both memory states for every model | 91–100% | held |
| PA1 | Prompt variants (499d317) | The errors-noted prompt lowers relay on the contradicted protein cell by at least 20 points for every model | DeepSeek 100% to 55%; Opus 5 and Argon from 18% and 14% to 0% and 2% | partly held |
| PA2 | Prompt variants | Removing the tool instruction changes relay on the protein and agreement cells by fewer than 10 points | Agreement cell fell 44, 29 and 57 points (Opus 5, Argon, Flash) | failed (P) |
| PA3 | Prompt variants | The two-fields format lowers relay on the protein cell by at least 20 points and states the database value faithfully in at least 90% of runs | First part DeepSeek only; value matched the record in 0–28% of runs | failed (P) |
| PA4 | Prompt variants | No variant lowers relay of coherent errors below 80% | Instruction variants 92–100%; two-fields 52% and 0% (Opus 5, Argon) | partly held |
| RC1 | Repetition (499d317) | A verbatim agreeing copy raises relay | Lowered: 68% to 52% (two-sided $p < 10^{-10}$) | failed (V) |
| RC2 | Repetition | A derived agreeing field raises relay more than a copy, by at least 10 points pooled | 73% against 52%, because the copy lowered relay | held |
| RC3 | Repetition | A disagreeing copy lowers relay | 0–3% against 57–86% | held |
| BH1 | Withholding (499d317) | Numeric relay at most 5% for every model | 0 of 709 | held |
| BH2 | Withholding | Identifier relay is unchanged from the annotate arm | Proteins 0–8% against 0–1%; compound names 4–48% against 40–92% (record formats differ) | partly held |
| IS1 | Identity (9234013) | Superseded values are relayed in at least 90% of anonymous runs | 97–100% | held |
| IS2 | Identity | Plausible relay without identity lies within 15 points of the full record | Argon and DeepSeek within; Opus 5 68% against 4% | partly held |
| IS3 | Identity | Impossible relay without identity is at most 20% | 0–3% | held |
| IS4 | Identity | Clean accuracy without identity is at least 90% | 100% | held |
| IS5 | Identity, addendum (c4f1f43) | Opus 5 with identity but no annotation lies within 15 points of the full or the anonymous record | 28% against 4% and 68% | failed (P) |

| ID | Experiment | Prediction and test | Outcome | Status |
|---|---|---|---|---|
| TH1 | Reasoning (9234013) | Without thinking, relay of contradicted values rises by at least 10 points | Opus 5 plausible 8% to 46% ($p = 2 \times 10^{-6}$); impossible 0% to 2%; Sonnet 5 fell | partly held |
| TH2 | Reasoning | Without thinking, plausible summary-record relay changes by fewer than 10 points | 92% to 94%; 100% to 100% | held |
| TH3 | Reasoning, addendum (cfbf0a7) | Argon at low reasoning effort relays contradicted plausible values at least 10 points more (one-sided McNemar) | 6% to 14% ($p = 0.06$) | failed (P) |
| NC1 | No-concerns prompt (9234013) | Relay changes by fewer than 10 points in every cell | 7 of 12 cells changed more (six rose by 11–43 points, one fell by 23) | failed (P) |
| NC2 | No-concerns prompt | Unprompted doubt falls below half of solicited flags | 6 of 12 cells | partly held |
| KW | No-concerns prompt | The keyword detector agrees with hand labels on 50 runs | $\kappa = 0.39$ (13 of 20 flags missed); replaced by a classifier ($\kappa = 0.84$ on these labels, 0.90 on 100 held-out runs labelled independently) | failed (D) |
| FD1 | Familiar field (9234013) | A familiar copy raises relay over the unfamiliar copy by at least 10 points, pooled | +9 points ($p = 0.015$); Opus 5 +13, Argon +15 | failed (V) |
| FD2 | Familiar field | A familiar copy does not raise relay over the weight alone by 10 points | 149 against 148 of 234 | held |
| FD3 | Familiar field | A familiar disagreeing copy lowers relay | 21 against 149 of 236 | held |
| CL | Superseded, addendum (cfbf0a7) | Opus 5 clean accuracy on the 79 superseded-record tasks is at least 95% | 79 of 79 | held |

Supplementary Table 22: Held-out validation of the unprompted-doubt classifier: 100 runs not used in its development (60 without the concerns field, 40 under the registered prompt), labelled by an independent annotator blind to the classifier, whose answers were committed before labelling (commit 6964e2e). Cohen's $\kappa$ 0.90 (95% bootstrap interval 0.80–0.98).

| Prompt | Runs | Agree | Classifier only | Annotator only | $\kappa$ | Annotator flags | Classifier flags |
|---|---|---|---|---|---|---|---|
| no concerns | 60 | 56 | 4 | 0 | 0.87 | 27/60 | 31/60 |
| registered | 40 | 39 | 1 | 0 | 0.93 | 30/40 | 31/40 |
| all | 100 | 95 | 5 | 0 | 0.90 | 57/100 | 62/100 |